THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 105	Complete graph on n points: $\frac{n(n-1)}{2}$.
2	**(a)** 37	Try triangular numbers.
3	**(b)** 800	Subtract the first 10 odd numbers, whose sum is 100.
4	**(b)** 39	$20^2 - 19^2$ is the next odd number.
5	**(b)** 11	Divide the total increase by the number of steps, which is one less than the term number.
6	**(a)** 88	Add the triangular number.
7	**(d)** 2025	Square of the triangular number.
8	**(d)** 41	Each triangular number adds the next counting number.
9	**(a)** Both A and R are true and R is the correct explanation of A.	15 + 4 = 19, and R is the rule of the pattern.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	1+3+5+7+9 = 25, and 25 = 5 x 5 is a square, but R does not explain why the sum is 25.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Complete graph idea: each pair plays once (1); $\frac{14\times13}{2} = 91$ (1).	1+1
12	$n \times n = 961$ so $n = 31$ (1); 31th odd number $= 2 \times 31 - 1 = 61$ (1).	1+1
13	Sum of first 24 odd numbers $= 576$ and first 9 odd numbers $= 81$ (1); Required sum $= 495$ (1).	1+1
14	Complete graph idea: each pair plays once (1); $\frac{16\times15}{2} = 120$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$a + (n-1)d = 365$ gives $n - 1 = (365 - 14) \div 9$ (1); $n = 40$ (1); Term before it $= 356$ (1).	1+1+1
16	First 20 odd numbers sum to 400 (1); First 25 odd numbers sum to 625 (1); Required sum $= 225$ (1).	1+1+1
17	Row 6 has $1 + 2 + \cdots + 6 = 21$ cubes (1); First rows: 1+3+6+10 (1); Total in first 4 rows $= 20$ (1).	1+1+1
	OR: $a + (n-1)d = 331$ gives $n - 1 = (331 - 23) \div 7$ (1); $n = 45$ (1); Term before it $= 324$ (1).	1+1+1
18	First 25 odd numbers sum to 625 (1); First 31 odd numbers sum to 961 (1); Required sum $= 336$ (1).	1+1+1
	OR: Row 7 has $1 + 2 + \cdots + 7 = 28$ cubes (1); First rows: 1+3+6+10+15 (1); Total in first 5 rows $= 35$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 13 sticks (1). (ii) 3 sticks (1). (iii) $4 + 9 \times 3 = 31$ sticks (1). (iv) $40 - 4 = 36$ and $36 \div 3 = 12$ more squares, so $1 + 12 = 13$ squares (2).	1+1+1+2
20	Figure 15: dots $= 5 + 14 \times 9 = 131$ (1); For 176 dots: steps from figure 1 $= (176 - 5) \div 9 = 19$ (1); Figure number $= 20$ (1); Check: substitute back to get the same number of dots (2).	1+1+1+2
	OR: 15th term of A $= 3 + 14 \times 7 = 101$ (1); 15th term of B $= 40 + 14 \times 3 = 82$ (1); Larger: A (1); Difference $= 19$ (2).	1+1+1+2

Mistakes to avoid:

- Thinking the sum of the first n natural numbers is $n \times n$; that is the sum of the first n **odd** numbers.
- Guessing a rule from only two terms. Always check the rule on at least three terms.
- Mixing up the nth odd number $2n - 1$ with the nth even number $2n$.
- Treating $2, 4, 8, 16$ as having a common difference; the gaps are $2, 4, 8$, so no constant is added.
- In the Virahanka sequence, adding the wrong two terms (the rule uses the *two terms just before*).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 1000	Subtract the first 15 odd numbers, whose sum is 225.
2	**(c)** 49	$25^2 - 24^2$ is the next odd number.
3	**(c)** 13	Divide the total increase by the number of steps, which is one less than the term number.
4	**(a)** 145	Add the triangular number.
5	**(a)** 3025	Square of the triangular number.
6	**(a)** 57	Each triangular number adds the next counting number.
7	**(c)** 190	Complete graph on n points: $\frac{n(n-1)}{2}$.
8	**(d)** 52	Try triangular numbers.
9	**(c)** A is true but R is false.	1+2+...+10 = 10 x 11 / 2 = 55, so A is true. The sum of the first n natural numbers is $\frac{n(n+1)}{2}$, not $n \times n$, so R is false.
10	**(d)** A is false but R is true.	The differences are 2, 4, 8, which are not equal, so A is false. R is true.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$n \times n = 1444$ so $n = 38$ (1); 38th odd number $= 2 \times 38 - 1 = 75$ (1).	1+1
12	Sum of first 28 odd numbers $= 784$ and first 11 odd numbers $= 121$ (1); Required sum $= 663$ (1).	1+1
13	Complete graph idea: each pair plays once (1); $\frac{17\times16}{2} = 136$ (1).	1+1
14	$n \times n = 1849$ so $n = 43$ (1); 43th odd number $= 2 \times 43 - 1 = 85$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$a + (n-1)d = 409$ gives $n - 1 = (409 - 6) \div 13$ (1); $n = 32$ (1); Term before it $= 396$ (1).	1+1+1
16	First 30 odd numbers sum to 900 (1); First 38 odd numbers sum to 1444 (1); Required sum $= 544$ (1).	1+1+1
17	Row 8 has $1 + 2 + \cdots + 8 = 36$ cubes (1); First rows: 1+3+6+10 (1); Total in first 4 rows $= 20$ (1).	1+1+1
	OR: $a + (n-1)d = 355$ gives $n - 1 = (355 - 31) \div 12$ (1); $n = 28$ (1); Term before it $= 343$ (1).	1+1+1
18	First 35 odd numbers sum to 1225 (1); First 42 odd numbers sum to 1764 (1); Required sum $= 539$ (1).	1+1+1
	OR: Row 9 has $1 + 2 + \cdots + 9 = 45$ cubes (1); First rows: 1+3+6+10+15 (1); Total in first 5 rows $= 35$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $1 + 2 + 3 + 4 + 5 + 6 = 21$ (1). (ii) $1 + 2 + \cdots + 8 = 36$ (1). (iii) 10 oranges (1). (iv) 45 oranges make a full stack of 9 rows ($1 + 2 + \cdots + 9 = 45$) (1); $50 - 45 = 5$ oranges are left (1).	1+1+1+2
20	Figure 15: dots $= 3 + 14 \times 11 = 157$ (1); For 212 dots: steps from figure 1 $= (212 - 3) \div 11 = 19$ (1); Figure number $= 20$ (1); Check: substitute back to get the same number of dots (2).	1+1+1+2
	OR: 14th term of A $= 11 + 13 \times 6 = 89$ (1); 14th term of B $= 6 + 13 \times 8 = 110$ (1); Larger: B (1); Difference $= 21$ (2).	1+1+1+2

Mistakes to avoid:

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- Guessing a rule from only two terms. Always check the rule on at least three terms.
- Mixing up the nth odd number $2n - 1$ with the nth even number $2n$.
- Treating $2, 4, 8, 16$ as having a common difference; the gaps are $2, 4, 8$, so no constant is added.
- In the Virahanka sequence, adding the wrong two terms (the rule uses the *two terms just before*).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 8	Divide the total increase by the number of steps, which is one less than the term number.
2	**(c)** 178	Add the triangular number.
3	**(d)** 4356	Square of the triangular number.
4	**(d)** 68	Each triangular number adds the next counting number.
5	**(a)** 300	Complete graph on n points: $\frac{n(n-1)}{2}$.
6	**(a)** 63	Try triangular numbers.
7	**(c)** 2100	Subtract the first 20 odd numbers, whose sum is 400.
8	**(d)** 69	$35^2 - 34^2$ is the next odd number.
9	**(a)** Both A and R are true and R is the correct explanation of A.	8 + 13 = 21, and R is the rule that builds the sequence.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	36 = 6 x 6 and 36 = 1+2+...+8, so A is true. R is true but only explains the square part.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Sum of first 33 odd numbers $= 1089$ and first 13 odd numbers $= 169$ (1); Required sum $= 920$ (1).	1+1
12	Complete graph idea: each pair plays once (1); $\frac{19\times18}{2} = 171$ (1).	1+1
13	$n \times n = 3025$ so $n = 55$ (1); 55th odd number $= 2 \times 55 - 1 = 109$ (1).	1+1
14	Sum of first 41 odd numbers $= 1681$ and first 16 odd numbers $= 256$ (1); Required sum $= 1425$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$a + (n-1)d = 425$ gives $n - 1 = (425 - 17) \div 8$ (1); $n = 52$ (1); Term before it $= 417$ (1).	1+1+1
16	First 40 odd numbers sum to 1600 (1); First 50 odd numbers sum to 2500 (1); Required sum $= 900$ (1).	1+1+1
17	Row 10 has $1 + 2 + \cdots + 10 = 55$ cubes (1); First rows: 1+3+6+10+15+21 (1); Total in first 6 rows $= 56$ (1).	1+1+1
	OR: $a + (n-1)d = 494$ gives $n - 1 = (494 - 4) \div 14$ (1); $n = 36$ (1); Term before it $= 480$ (1).	1+1+1
18	First 45 odd numbers sum to 2025 (1); First 54 odd numbers sum to 2916 (1); Required sum $= 891$ (1).	1+1+1
	OR: Row 12 has $1 + 2 + \cdots + 12 = 78$ cubes (1); First rows: 1+3+6+10+15+21 (1); Total in first 6 rows $= 56$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 25 tiles (1). (ii) $36 - 25 = 11$ tiles (1). (iii) 11 is the 6th odd number (1). (iv) The sum of the first 9 odd numbers is $9 \times 9 = 81$ (1); she adds the 10th odd number, 19 tiles ($100 - 81 = 19$) (1).	1+1+1+2
20	Figure 15: dots $= 6 + 14 \times 8 = 118$ (1); For 158 dots: steps from figure 1 $= (158 - 6) \div 8 = 19$ (1); Figure number $= 20$ (1); Check: substitute back to get the same number of dots (2).	1+1+1+2
	OR: 13th term of A $= 9 + 12 \times 9 = 117$ (1); 13th term of B $= 30 + 12 \times 5 = 90$ (1); Larger: A (1); Difference $= 27$ (2).	1+1+1+2

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